[Abstract]

RoFormer: Enhanced Transformer with Rotary Position Embedding

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Abstract: Position encoding recently has shown to be effective in the Transformer architecture. It enables valuable supervision for dependency modeling between elements at different positions of the sequence. We propose a novel method named Rotary Position Embedding (RoPE) to effectively leverage the positional information.

Specifically, the proposed RoPE encodes the absolute position with a rotation matrix and meanwhile incorporates the explicit relative position dependency in self-attention formulation. Notably, RoPE enables valuable properties, including the flexibility of sequence length, decaying inter-token dependency with increasing relative distances, and the capability of equipping the linear self-attention with relative position encoding.

[Rotary Position Embedding Formulation]

3.2 Rotary Position Embedding (RoPE): Unlike the additive sinusoidal positional encoding introduced in Vaswani et al. (2017) or the learned absolute position embeddings used in Devlin et al. (2019) BERT, RoPE multiplies the query and key representations by an orthogonal rotation matrix R_(Theta, m)^d parameterized by token position index m.

Because the inner product q_m^T k_n depends solely on the relative distance (m - n) and the token representations, RoPE naturally injects relative position awareness directly into Self-Attention without altering value vectors or requiring O(N^2) relative position bias tables.

[Long-Context Extrapolation & Experiments]

4 Experiments: Evaluating RoFormer on WMT 2014 English-to-German translation (achieving 27.5 BLEU on Transformer base) and pre-training on BERT downstream GLUE benchmarks demonstrates faster convergence and superior long-sequence extrapolation beyond 512 tokens compared to absolute positional encodings.